

Indian Institute of Technology Guwahati
Department of Electronics and Electrical Engineering
EE101 - Basic Electronics

Solution: Pre-Tutorial Assignment - 7

Date: Oct 5, 2023

Marks: 2

1. Find the sinusoids represented by these phasors:

(a) $\bar{I} = -3 + j4$ A

(b) $\bar{V} = j8e^{-j20^\circ}$ V

Solution:

(a) For $\bar{I} = -3 + j4 = 5\angle 126.87^\circ$.

Transforming this to the time domain gives

$$i(t) = 5 \cos(\omega t + 126.87^\circ) \text{ A.}$$

(b) For $j = 1\angle 90^\circ$, we have

$$\begin{aligned}\bar{V} &= j8e^{-j20^\circ} \\ &= (1\angle 90^\circ)(8\angle -20^\circ) \\ &= 8\angle (90^\circ - 20^\circ) \\ &= 8\angle 70^\circ \text{ V.}\end{aligned}$$

Converting this to the time domain gives

$$v(t) = 8 \cos(\omega t + 70^\circ) \text{ V.}$$